MedAgentBench: Non-LLM Routers vs Jonathan’s LLM

Same replay views, outcomes and costs · Patient-disjoint split: 90 train / 60 validation / 150 test · Benchmark pass/fail accuracy

T1: decide after the full weak attempt | T3: decide at round 2 | Executors: Qwen2.5-7B (weak) / Qwen2.5-72B-AWQ (strong)

PUBLISHED LLM · NO RISK GATE T1 61.33% accuracy / 54.11% cost T3 63.33% accuracy / 81.47% cost

MAIN TABLE: same high-risk escalation gate for every router · Seed 17 · Non-LLM thresholds chosen on validation; Jonathan uses θ = 0.50

Router	Training	How the model works	Full attempt · T1		Round 2 · T3		T3 accuracy range
			Acc. %	Cost %	Acc. %	Cost %	seeds 0 / 1 / 2
Jonathan LLM	**GRPO (A1)**	Qwen2.5-1.5B + LoRA; pretrained backbone.	**61.33**	**80.99**	**63.33**	**81.47**	Not reported
Linear Softmax	**GRPO (A1)**	Linear softmax on TF-IDF + view statistics.	44.00	71.72	63.33	81.47	53.33–56.67
MLP · 64	**GRPO (A1)**	One ReLU hidden layer with 64 units.	61.33	80.99	63.33	81.47	54.00–62.67
MLP · 128 → 64	**GRPO (A1)**	Two ReLU hidden layers: 128 then 64 units.	61.33	80.99	61.33	71.39	54.00–63.33
Random Forest	**Supervised**	Average 300 bootstrapped decision trees.	61.33	80.99	63.33	81.47	63.33–63.33
Extra Trees	**Supervised**	Average 300 trees with randomized splits.	61.33	80.99	63.33	81.47	63.33–63.33
Decision Tree	**Supervised**	One tree of learned feature-split rules.	61.33	80.99	61.33	71.39	61.33–61.33
Linear SVM	**Supervised**	Maximum-margin linear decision boundary.	61.33	81.58	**63.33**	**77.68**	63.33–63.33
RBF SVM	**Supervised**	Maximum-margin boundary with an RBF kernel.	61.33	81.58	**63.33**	**77.68**	63.33–63.33
Logistic Regression	**Supervised**	Linear classifier fitted by weighted logistic loss.	61.33	81.58	**63.33**	**77.68**	63.33–63.33
Naive Bayes	**Supervised**	Class-conditional token likelihoods; text only.	61.33	80.99	63.33	81.47	63.33–63.33

Cost denominator: full weak + strong on the same 150 tasks = 100%. Full always-escalate: 54.00% accuracy; two-action oracle: 63.33%.

GRPO (A1) · neural classifiers

Linear / MLP: policy-gradient training, as in the LLM recipe.

Fresh classifier; sample 8 actions per task.

Advantage = reward − group mean; entropy bonus 0.03.

40 epochs · lr 1e−5 · accumulation 8 · gradient clip 1.

No std normalization, KL term or PPO clipping.

Best checkpoint selected by validation delivered reward.

Supervised · classical classifiers

Label = argmax(R_W, R_S); weight = |R_W − R_S|.

Fit the classifier to reward-optimal routing actions.

Hyperparameters and thresholds selected on validation.

SVM: patient-grouped out-of-fold score calibration.

This is supervised training, not GRPO.

Same view features; Naive Bayes uses text only.

Shared objective and readout

R = Q − 0.1 × (C / c_s) − max(0, C / c_s − 1).

Q is the benchmark’s pass/fail, not QA lenient match.

T1 costs: deliver c_w; escalate c_w + c_s.

T3 costs: deliver c_w; escalate c_w2 + c_s.

Common gate: τ ≥ 0.8 forces escalation.

Inputs: task text/context + observed weak attempt.

SVM / logistic regression: 63.33% accuracy at 77.68% cost

Same success rate as Jonathan’s T3; 3.79 cost points less. Repeats 0/1/2 agree.

Why? Avoid 15 ineffective strong upgrades.

Both models fail on those task7 cases; failures remain.

Lower compute does not mean improved clinical outcomes.

Main table applies the same cached τ ≥ 0.8 gate to all rows. It changes the LLM’s T1 cost from 54.11% to 80.99%; T3 is unchanged. No test-based threshold selection.

Routers actually run; executor outcomes are replayed, not rerun. Cost excludes router / risk-scorer compute. Previously inspected test; small, homogeneous task types; GRPO repeats vary.